

Signal Analysis Report — Full Pipeline Findings

Input: `CommSignal2_demod_train_0042.sigmf-data` · format `cf32` · 40960 samples

1. Input Recording

Property	Value
Source file	<code>CommSignal2_demod_train_0042.sigmf-data</code>
Raw IQ format	<code>cf32</code> (complex float32)
Recording length	40960 complex samples (≈ 1.64 ms)
Sample rate	25.00 MHz
Center frequency	2.437 GHz

A short burst of real over-the-air radio from the MIT RF Challenge dataset, known ground truth: QPSK.

2. Detection — What the Spectrum Shows

14.9 dB SNR · psd_ratio	732 kHz Bandwidth · 3 dB	-88.0 dBFS Noise floor · MAD	4 Bursts · segments
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Measurement	Value	What this means
Signal strength (SNR)	14.96 dB	moderately strong — clearly above the noise
Signal power	-73.1 dBFS	
Noise floor	-88.0 dBFS	quiet recording background
Bandwidth (-3 dB)	732.4 kHz	narrow-band signal, occupies only $\approx 2.9\%$ of the 25 MHz recording band
Energy bursts detected	4	the transmission arrives as several short bursts

So: a real, moderately strong, narrow-band digital signal arriving in bursts — not a wall of noise and not a single continuous tone.

3. Characterization — The Signal's Shape

Feature	Value	What this means
4th-order cumulant ($C_{42\text{norm}}$)	0.41	≈ 0.41 — between QPSK (1.0) and 16QAM (0.68) expectations
Rotational symmetry ($C_{21\text{sym}}$)	0.010	≈ 0.00 — strong 4-fold phase symmetry (QPSK signature)
Peak-to-average ratio (PAPR)	7.4 dB	≈ 7.4 dB — multi-level style, not a constant-envelope tone
Amplitude kurtosis	-0.61	-0.61 — single amplitude shell (4 phase states, one ring)
Frequency spread (normalized)	0.105	0.105 — mostly phase modulation, not tone shifting (not FSK)
Spectral flatness	0.045	0.045 — peaked spectrum: a real tone-rich signal, not noise

The combination — one amplitude ring plus four symmetric phase states — is the classic footprint of QPSK.

3.1 Derived Parameters

Parameter	Value	Confidence	Method
Symbol rate	11.00 MHz	0.90	envelope
Frequency offset (CFO)	-561.5 kHz	0.90	psd_peak

⚠ **Two readings should not be trusted.**

- The **symbol-rate estimate (~11 MHz)** is far faster than the measured 732 kHz bandwidth allows — a sanity check on bandwidth did not fire. Ignored for the verdict.
- The **CFO estimate used the PSD-peak method**, which is only valid for narrow continuous tones, not a wide QPSK waveform. The demodulator still locked on using its own carrier recovery, so this number is not relied on.

4. Modulation Classification — Ranked Candidates

Rank	Candidate	Score	Confidence	Verdict
1	QPSK	0.411	0.907	WINNER
2	16-QAM	0.377	0.832	close runner-up (multi-ring shape missing)
3	BPSK	0.101	0.222	no — 2-fold symmetry not seen
4	2-FSK	0.057	0.125	no — not a tone-shifter
5	4-FSK	0.051	0.113	no — constant envelope absent, spectrum too tight
6	Noise / CW	0.004	0.050	no — clearly a modulated signal present

Classifier WHY: C42 = 0.41 (QPSK target ≈ 1.0 , still close enough once symmetry dominates), 4-fold symmetry $C21_{\text{sym}} = 0.01$ (< 0.25), single-shell amplitude kurtosis = -0.61 (< -0.40), PAPR = 7.4 dB.

5. Demodulation — Pulling Symbols Out of the Waveform

Property	Value	What this means
Demodulated as	QPSK	assumed QPSK for slicing
Symbols recovered	2554	≈ 2550 symbols from the bursts
EVM (RMS)	45.0% (-6.9 dB)	poor — 45% is far from the clean $< 10\%$ you expect
EVM (peak)	110%	$> 100\%$ means some symbols land near the wrong point entirely
Samples per symbol	16.0	≈ 16 — timing recovered
Timing offset	6.93 samples	
Residual phase error	0.000 rad	≈ 0 — carrier lock achieved
Bits produced	5108	raw candidate bits for decoding

Carrier lock was clean, but symbol scatter is high. On real noisy hardware this is expected — but it means the recovered bits are too corrupted to decode into a valid frame (next section).

6. Decoding — Did Any Data Frame Verify?

#	Decode chain	Status	Viterbi errs	RS corr	CRC
1	viterbi block(1, 3) preamble+crc16	FAILED	0	0	none
2	viterbi block(1, 3) preamble+crc32	FAILED	0	0	none
3	rs255_223 block(1, 3) preamble+crc16	FAILED	0	0	none
4	rs255_223 block(1, 3) preamble+crc32	FAILED	0	0	none
5	viterbi+rs255_223 block(1, 3) preamble+crc16	FAILED	0	0	none
6	viterbi+rs255_223 block(1, 3) preamble+crc32	FAILED	0	0	none

Every candidate chain failed its checksum — no data frame could be verified. This is reported honestly: the assistant will not invent a decode when the bits do not support one.

7. Evidence Graph — Cross-Stage Scoring

Rank	Kind	Candidate	Score	Quality	Evidence items
1	modulation	16QAM	0.695	fair	9
2	modulation	QPSK	0.680	fair	13
3	modulation	2-FSK	0.560	fair	5
4	modulation	4-FSK	0.558	fair	5
5	modulation	BPSK	0.537	fair	7

6	decode_chain	viterbi block(1, 3) preamble+crc16	0.509	fair	5
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The evidence graph sees 16-QAM and QPSK as essentially tied. QCAM has more direct evidence items; QPSK leads on classification. This is the honest state of the evidence before the full chain search.

8. Hypothesis Engine — Ranked Full Chains

Evaluated **10** candidate chains in 274 s (live demodulation and decoding ran for each group).

#	Modulation	sps	Deinterleaver	FEC	Frame	Composite score	Uncertainty
1	QPSK	1.0	block(4,8)	none	preamble+crc16	0.522	0.468
2	QPSK	1.0	block(4,8)	none	preamble+crc32	0.522	0.468
3	QPSK	1.0	block(4,8)	rs255_223	preamble+crc16	0.522	0.468
4	QPSK	1.0	block(4,8)	rs255_223	preamble+crc32	0.522	0.468
5	QPSK	1.0	block(4,8)	rs255_239	preamble+crc16	0.522	0.468
6	QPSK	1.0	block(4,8)	rs255_239	preamble+crc32	0.522	0.468
7	QPSK	1.0	block(8,16)	none	preamble+crc16	0.522	0.468
8	QPSK	1.0	block(8,16)	none	preamble+crc32	0.522	0.468

8.1 Why the Top Hypothesis Won

QPSK chain — composite **0.522** (modulation 0.66 · demod 0.40 · decode 0.40)

Modulation evidence: classifier QPSK score 0.411 (supports), C42 fit 0.74, PAPR fit 0.98, kurtosis fit 1.00, top-candidate margin 0.034.

Demodulation evidence: EVM ≈ 57% (poor), residual phase ≈ 0 rad.

Decode evidence: CRC **FAIL**, Viterbi errs 0, RS corr 0, preamble correlation 1.000 (locked at bit 29327).

Context: SNR = 15.0 dB. Uncertainty: std ≈ 0.47 — a modest-confidence call, not a certainty.

8.2 Alternatives Considered

- Chain[1aba3757]: 16QAM -> 16QAM | sps=1.00 -> none -> none -> preamble+crc16 | score=0.513
- Chain[cfe52b1e]: 16QAM -> 16QAM | sps=1.00 -> none -> none -> preamble+crc32 | score=0.513
- Chain[8215e1ea]: 16QAM -> 16QAM | sps=1.00 -> none -> rs255_223 -> preamble+crc16 | score=0.513

The 16-QAM chains scored 0.513 — just below QPSK. The engine kept them and disclosed them rather than hiding the near-tie.

9. Bottom Line

- The recording is a real, moderately strong, narrow-band digital signal arriving in four short bursts.
- Its dominant traits — one amplitude ring, strong 4-fold phase symmetry, peaked spectrum — classify it as **QPSK** with ≈ 0.9 confidence; 16-QAM is the close runner-up.
- The demodulator locked on (clean phase, ≈ 0 residual error) and pulled about **2 550 symbols**, but symbol quality was poor (EVM ≈ 45%).
- Because the symbols are so scattered, **no decode chain could verify a data frame** — reported honestly.
- The hypothesis engine ranks the QPSK chain first (composite 0.522) with the 16-QAM chains as disclosed alternatives, and attaches an explicit uncertainty (std ≈ 0.47).
- Two auto-measurements (symbol rate ≈ 11 MHz, PSD-peak CFO) were internally inconsistent and are flagged as untrustworthy.

Net answer: **confident QPSK identification, honest failure on full decoding**, with every value and every caveat reported above.